Statement of Work for Project “”

Title and Authors

- Title: *Team of AI Stock Experts: A Multi-Agent System for Stock Recommendations*
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Background and Motivation

Most Americans (62%) own stocks (ref), but beginners often struggle with confident, data-driven decisions due to information overload, lack of personalized guidance, and limited access to advanced tools like hedge fund models. They face challenges in selecting stocks, assessing risks, managing emotions, interpreting macroeconomic data, and experimenting with new strategies. Even experienced investors find market analysis complex, requiring integration of fundamental metrics, technical indicators, and broader market conditions. Manually combining these is time-consuming, bias-prone, and demands expertise. Current tools provide fragmented insights, highlighting the need for an intelligent system delivering tailored stock recommendations based on user risk profiles and goals.

Problem Statement

Develop an AI-driven multi-agent application that integrates diverse financial data and expert-inspired reasoning to generate stock recommendations tailored to individual investors’ needs.

Data Sources

Source & Description:

Source	Format	Description
Yahoo Finance (yfinance)	Time series	Stock data Open/High/Low/Close/Volume
Financial Modeling Prep (FMP API) https://site.financialmodelingprep.com/developer/docs	Tabular + Text (API)	Earnings, fundamental metrics, Market data
BLS website (https://www.bls.gov/)	Text	Macroeconomic data such as GDP, unemployment, interest rates, etc
Books / Blogs (For RAG training)	Text	Aswath Damodaran's Valuation Guides (https://pages.stern.nyu.edu/~adamodar/) Machine Learning in Finance (book) https://www.investopedia.com/ https://www.investor.gov/introduction-investing

Key Attributes

Attribute Type	Key Features
Price & Returns	Daily open, high, low, close, volume; percent change; cumulative return; volatility (yfinance)
Fundamental Metrics	P/E ratio, P/B ratio, ROE, ROA, Debt/Equity, EPS, EBITDA, net margin(FMP API)
Technical Indicators	3-month return, RSI, MACD, 50/200-day moving averages (derived from yfinance)
Macroeconomic Indicators	Interest rates, unemployment, inflation , GDP growth (FMP API/website)
Textual Knowledge (RAG)	Definitions, investment strategies, valuation logic

Relevance to the Project

The selected datasets are directly aligned with our goal: to build a **multi-agent system** that integrates multiple sources of financial data and expert reasoning to deliver **personalized and explainable stock recommendations**.

- **Price and volume data** enable the calculation of momentum indicators, volatility measures, and trend signals, which are essential for short- and medium-term decision-making.

- **Fundamental metrics** support the evaluation of companies based on financial health, valuation, and earnings performance, which are key for long-term investment considerations.

- **Macroeconomic data** provides broader economic context—such as interest rate trends, employment levels, and inflation—that influences market cycles and asset class performance.

- **RAG training documents**, such as valuation textbooks and trusted finance blogs, equip the system with expert-level knowledge and vocabulary, allowing it to deliver rational, explainable recommendations in natural language.

- **User-specific inputs**—such as risk appetite, investment horizon, and sector preferences—ensure that the recommendations align with individual goals, constraints, and decision styles.

Together, this diverse and rich data foundation supports the construction of a robust and customizable recommendation system, capable of integrating multiple financial viewpoints while delivering insights that are actionable and align with real investor needs.

Data Quality Concerns & Mitigation

We anticipate several data quality challenges and have planned mitigation strategies accordingly. To address **missing data**, especially among small-cap stocks, we'll initially limit our universe to the S&P 500 or large-cap ETFs, ensuring consistent coverage. **API rate limits** from sources like Yahoo Finance and Financial Modeling Prep will be managed through caching and by testing with smaller subsets before scaling up.

Objectives

- Understand investor preferences and financial goals to devise an investment strategy
- Convert expert writings to build expert agents for different subtasks of the stock evaluation
- Collect relevant data sources that will help in understanding the fundamental and technical analysis of the company
- Recommend stocks with reasoning and share insights into the fundamentals of the company, as well as technical endpoints
- Provide tools/ML algorithms where needed, to help agents make necessary inferences
- Generate recommended stock reports from the front-end interface

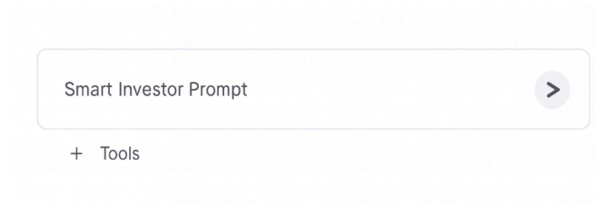
Learning Emphasis

The project will use LLMs, AI Agents, RAG, LLM fine-tuning (if needed), quant finance knowledge and MCP.

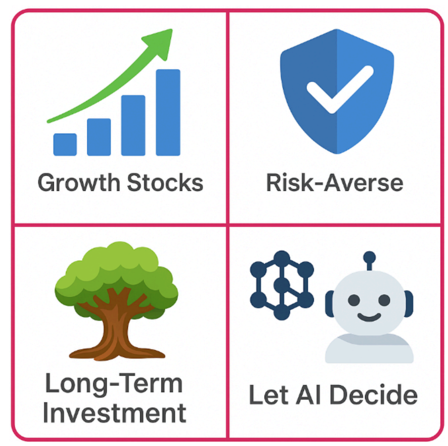
Application Mock Design

The application will feature two main interfaces:

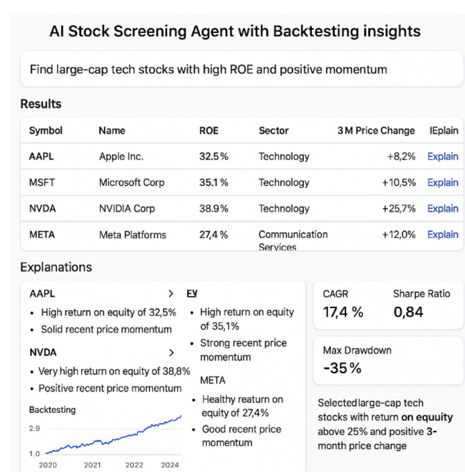
- Chat Interface: A conversational interface where investors describe their goals, preferences (risk, horizon, sectors), and the system gathers necessary inputs.



- Stock Agents: A window with icons showing existing and newly created investing tools for stock analysis



- Stock Reports: For each investing agent, list of reports generated, recommending stocks and explaining the rationale for recommendation



AI Stock Screening Agent with Backtesting insights

Find large-cap tech stocks with high ROE and positive momentum

Results

Symbol	Name	ROE	Sector	3 M Price Change	IEplain
AAPL	Apple Inc.	32.5 %	Technology	+8,2%	Explain
MSFT	Microsoft Corp	35.1 %	Technology	+10,5%	Explain
NVDA	NVIDIA Corp	38.9%	Technology	+25,7%	Explain
META	Meta Platforms	27,4 %	Communication Services	+12,0%	Explain

Explanations

AAPL
- High return on equity of 32,5%
- Solid recent price momentum

NVDA
- Very high return on equity of 38,8%
- Positive recent price momentum

Backtesting

2.9
1.0
2020 2021 2022 2024

EV
- High return on equity of 35,1%
- Strong recent price momentum

META
- Healthy reaturn on equity of 27,4%
- Good recent price momentum

CAGR 17,4 %
Sharpe Ratio 0,84

Max Drawdown -35 %

Selectedlarge-cap tech stocks with return **on equuity** above 25% and positive **3-** month price change

Research and Development

- We'll review RAG techniques and, if needed, fine-tune Large Language Models
- Dynamic creation of Agents based upon needs through an orchestrator
- Explore investment methodologies and generate tools to assist Agents in making decisions
- MCP server integration and Agent-to-Agent Communication (if needed)

Fun Factor

The project is essentially a "dream team of AI investors": each agent acts like a mini-expert (value investor, quant trader, risk manager), collaborating to produce results.

Limitations and Risks

Possible challenges in obtaining a large and diverse enough dataset.
Generating RAGs from expert writings, containing tabular data and snapshots of stock trends
Limited availability of tools to assist agents in making decisions
Copyright issues for expert text

Milestones

- Stock Recommender High-Level Design - W40
- Data collection from different sources, like fundamental Data for stocks, Technical data of stock tickers and setup environment -W40
- Collecting user requirements by the orchestrator and asking additional questions for missing data -W41
- Breakdown of the requirement into different tasks -W41
- Model creation for screening and backtesting strategies -W40-41
- RAG generation from expert writings and evaluating performance for cases of tables and visual figures interpretations -W42
- Tool creation for Agentic assistance -W43
- MCP server setup for data sources _W43
- Agents/subagents creation and evaluation of performance -W44
- Agent-to-Agent communication -W44
- Front-end development for front-end interfaces - W45
- Connect agents, orchestrator, and UI via APIs; establish CI/CD and automated tests. -W46
- Generation of Reports -W47